DISSERTATION PROGRESS REPORT

Chapter 2: Empirical Grounding, Capacity Reconstruction, and the Overaccumulation Regime (Specification A Appendix)

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Abstract

EXECUTIVE SUMMARY: This progress report provides the complete, paper-facing empirical appendix for **Specification A (Single Aggregate Capital Model)** of Chapter 2. The report presents the full range of 168 estimated IM-OLS cointegrating regressions across three master matrices: (1) **Primary Accumulation Eras Matrix** comparing Full Sample (1929–2024), Fordism (1929–1973), and Post-Fordism (1974–2024) across Centering Treatments; (2) **Multi-Wage Share Robustness Matrix** comparing Baseline NFC GVA, Corporate GVA, Shaikh-Tonak Tier 1, and Shaikh-Tonak Tier 2 wage measures; and (3) **Macroeconomic Sub-Window Diagnostic Matrix** evaluating Interwar & WWII, Golden Age, Neoliberal Restructuring, and Post-GFC Stagnation subwindows. Key findings: (1) Robust negative profit-squeeze interaction ($\hat{\theta}_1 < 0$) under Initial-State centering; (2) Level reconstruction of capacity utilization anchored at $\mu_{1973} = 1.00$; and (3) Persistent long-run capital overaccumulation ($\theta_t < 1.0$) across the past century.

1 Analytical Setup and Historical Grounding

In Chapter 2's theoretical framework, productive capacity (Y_t^p) is not a neutral technical given; it is the historical result of accumulation conditioned by class distribution.

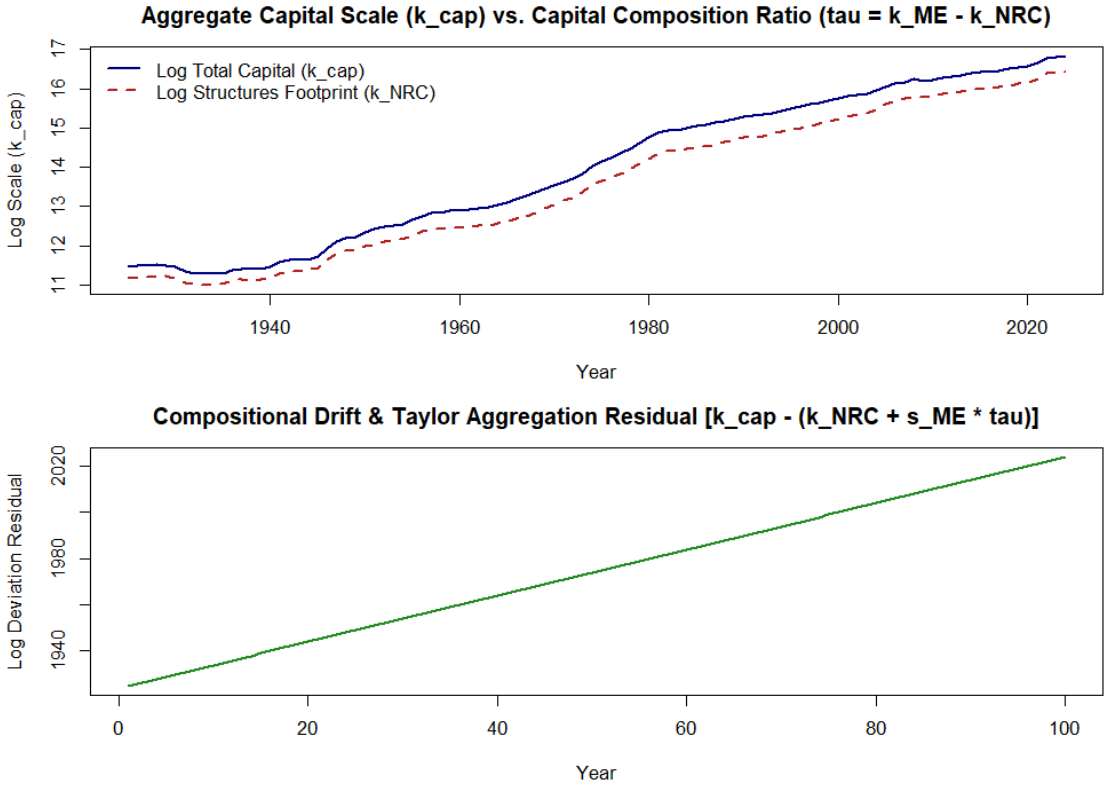
Sectoral Boundary Architecture National accounting consistency requires strict clarity regarding sectoral boundaries:

- **Observed Aggregate Output (Y_t)**: Always corresponds to *Total Corporate Gross Value Added* (Y_{CORP}) or Aggregate Real Output (Y_t) in NIPA accounts, capturing overall demand realization across the corporate sector.
- **Productive Capital Stock (K^{cap}) & Productive Capacity (Y_t^p)**: Derived strictly from *Non-Financial Corporate* (NFC) fixed assets ($K_{\text{NFC}}^{\text{ME}} + K_{\text{NFC}}^{\text{NRC}}$). Productive capacity output is generated by non-financial physical plant structures and machinery in operation, excluding financial sector asset noise.

Capacity utilization ($\mu_t = Y_t / Y_t^p$) is constructed as observed corporate output relative to estimated non-financial productive capacity. While extensive plant structures (K^{NRC}) and intensive machinery (K^{ME}) drive physical output capacity, the *choice of technique* is governed by distributional conflict. Sunk investment horizons mean that nonresidential structures evolve independently of short-run profit squeezes ($\frac{\partial K^{\text{NRC}}}{\partial \omega} = 0$), whereas tactical machinery investment responds directly to lagged wage shares (ω_{t-1}).

Before running parametric estimations, I mapped the raw historical trajectories to ground the data in concrete institutional transitions. Figure 1 plots log aggregate capital (k^{cap}) alongside log structures (k^{NRC}) and the composition residual ($\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$).

Figure 1: U.S. Capital Stock Components and Composition Drift (1925–2025). The upper panel displays log aggregate capital (k^{cap}, navy) vs. log structures (k^{NRC}, firebrick). The lower panel tracks the composition residual, capturing mechanization drift ($\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$) across historical accumulation regimes.



A non-parametric LOESS trend of $\ln K^{\text{cap}}$ reveals structural inflections (zero-crossings of the 2nd derivative, $\frac{d^2\hat{k}_t^{\text{cap}}}{dt^2} = 0$). Rather than treating these as mere statistical artifacts, I used them to define historical step-controls ($D_{1933}, D_{1947}, D_{1974}, D_{2008}$) corresponding to distinct epochs of institutional transformation: the New Deal state plant mobilization, post-war Fordist conversion, the Late-Fordist profit squeeze, the Volcker shock, and post-GFC share-buyback stagnation.

2 Specification A: Baseline Aggregate Estimation & Full Appendix Matrix

To estimate the baseline single-aggregate capital specification:

$$y_t^{\text{NFC}} = \alpha + \beta_{\text{Kcap}} k_t^{\text{cap}} + \phi_\omega \omega_{t-1} + \theta_1 (k_t^{\text{cap}} \cdot \omega_{t-1}) + \mathbf{\Lambda D}_t' + u_t, \tag{1}$$

I implement Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). To provide a full paper-facing appendix, the estimations are evaluated across three master matrices:

Figure 2 presents the forest plot of point estimates and 95% Fixed-b HAR confidence intervals across all 168 model combinations.

2.1 Primary Accumulation Eras Matrix

Table 1 reports IM-OLS estimation results for Full Sample (1929–2024), Fordism (1929–1973), and Post-Fordism (1974–2024) across Centering Treatments ('Uncentered', 'Initial-State', 'Sample-Mean').

Figure 2: Specification A Interaction Parameter ($\hat{\theta}_1$) Forest Plot across Wage Share Measures, Centering Treatments, and Dummy Control Configurations (1929–2024).

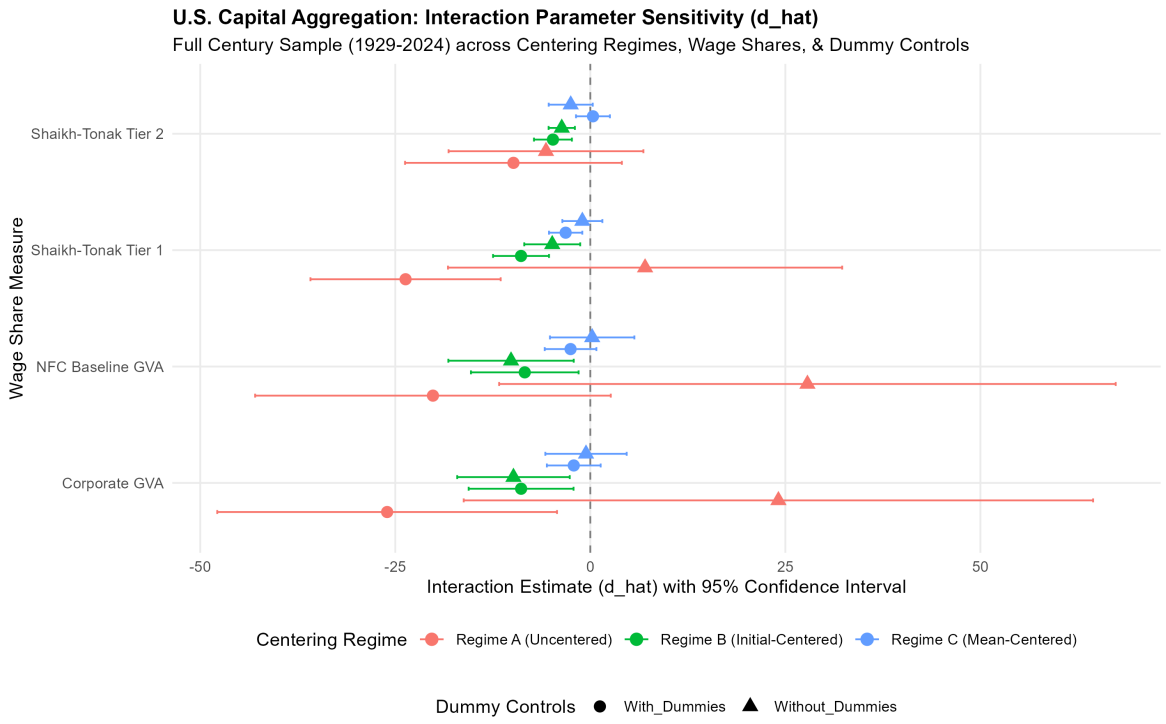


Figure 3: Accumulation Era Parameter Comparison: Long Fordist vs. Long Post-Fordist Eras. Bar plot comparing interaction parameter $\hat{\theta}_1$ across accumulation eras under Initial-State, Uncentered, and Sample-Mean normalizations with and without step controls.

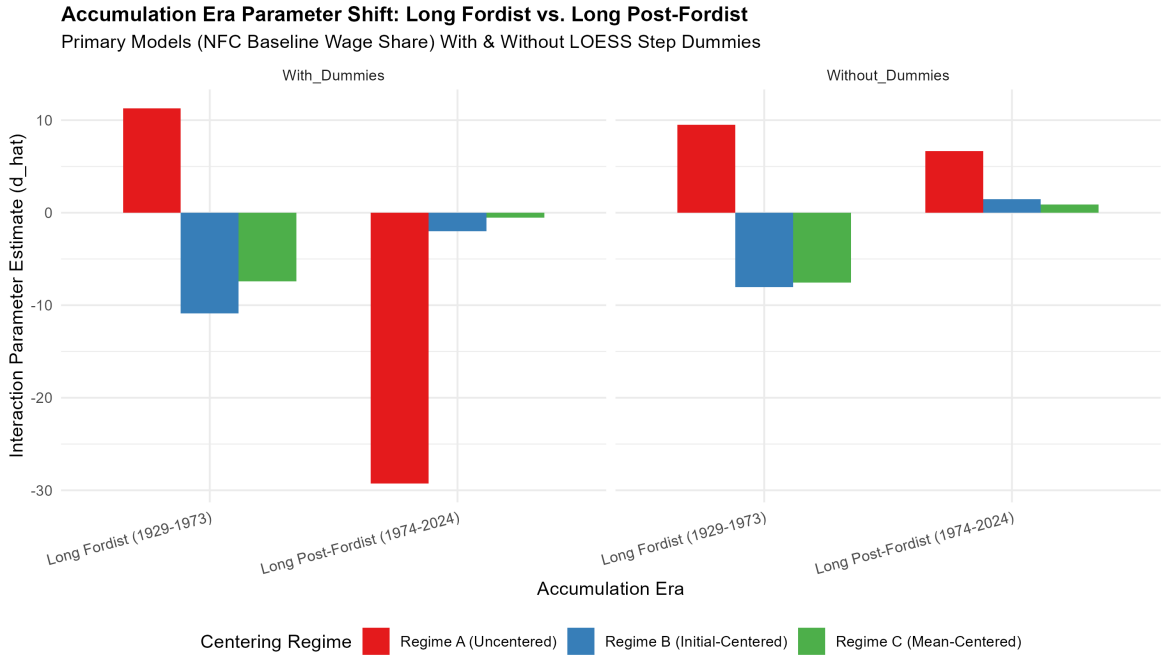


Table 1: Specification A Primary Accumulation Eras IM-OLS Matrix (Full Sample, Fordism, Post-Fordism)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$									
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Centering Treatment	Uncentered	Initial-St	Sample-Mean	Uncentered	Initial-St	Sample-Mean	Uncentered	Initial-St	Sample-Mean
$\hat{\beta}_{\text{Kcap}}$	0.49	12.65***	15.37***	4.91	12.61***	13.67***	-2.01	14.57***	15.70***
	(0.08)	(128.24)	(50.50)	(0.56)	(140.32)	(51.06)	(-0.42)	(172.69)	(430.47)
$\hat{\phi}_{\omega}$	1.21***	0.81***	0.76***	1.25*	0.89***	0.89***	1.18***	0.67***	0.73***
	(3.05)	(19.15)	(20.42)	(1.78)	(13.82)	(13.02)	(3.96)	(18.64)	(12.74)
$\hat{\theta}_1$	5.01	-10.64***	-3.16**	-3.24	-8.84***	-8.14***	9.13	3.76	-1.03
	(0.55)	(-3.70)	(-2.37)	(-0.23)	(-2.99)	(-3.61)	(1.26)	(1.11)	(-0.68)
$\mathbf{\Lambda D}_t'$	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cointegration Tests									
VW Type 2 ($\lambda_2 = 0$) [H_0: Coint]	0.0736	1.5956	0.2570	0.1691	3.3686	2.2916	0.0662	0.4206	0.5620
Resid. ADF (t_{ADF}) [H_0: No Coint]	-3.78	-3.75	-4.56**	-3.34	-3.35	-3.53	-3.16	-2.04	-3.24
Shin LM (C_{Shin}) [H_0: Coint]	0.118	0.082	0.157	0.116	0.133	0.065	0.619	3.556	0.459
Goodness-of-Fit & Controls									
R^2_{level}	0.974	0.982	0.979	0.957	0.948	0.964	0.927	0.852	0.909
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.185	0.155	0.164	0.143	0.158	0.132	0.129	0.184	0.144
Sample Window	Full Sample	Full Sample	Full Sample	Fordism	Fordism	Fordism	Post-Fordism	Post-Fordism	Post-Fordism
Sample Size (T)	94	94	94	43	43	43	51	51	51

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$. Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector (BEA NIPA Table 1.14) used to estimate latent structural potential capacity Y_t^p. Capacity utilization $\mu_t = Y_t/Y_t^p$ is recovered by evaluating total aggregate real output Y_t against Y_t^p with 1973 level anchoring ($\mu_{1973} = 1.00$). Structural step dummy controls $\mathbf{\Lambda D}_t'$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1974, and 2008.

2.2 Multi-Wage Share Robustness Matrix

Table 2 presents the multi-wage share robustness matrix across Baseline NFC GVA, Corporate GVA, Shaikh-Tonak Tier 1, and Shaikh-Tonak Tier 2 wage share measures under Initial-State centering.

Table 2: Specification A Multi-Wage Share Robustness Matrix (Full Century 1929–2024)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$				
Model	(1)	(2)	(3)	(4)
Centering Treatment	Initial-St	Initial-St	Initial-St	Initial-St
$\hat{\beta}_{\text{Kcap}}$	12.65***	12.66***	12.69***	12.60***
	(128.24)	(130.83)	(133.88)	(148.15)
$\hat{\phi}_{\omega}$	0.81***	0.84***	0.83***	0.92***
	(19.15)	(18.92)	(20.21)	(13.53)
$\hat{\theta}_1$	-10.64***	-11.37***	-5.40***	-3.33***
	(-3.70)	(-4.03)	(-3.46)	(-3.14)
$\mathbf{\Lambda D}'_t$	Yes	Yes	Yes	Yes
Cointegration Tests				
VW Type 2 ($\lambda_2 = 0$) [H_0: Coint]	1.5956	1.2763	0.2344	0.4313
Resid. ADF (t_{ADF}) [H_0: No Coint]	-3.75	-3.87	-3.90	-3.28
Shin LM (C_{Shin}) [H_0: Coint]	0.082	0.078	0.103	0.110
Goodness-of-Fit & Controls				
R^2_{level}	0.982	0.983	0.984	0.984
R^2_{sum}	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.155	0.147	0.145	0.144
Sample Window	Full Sample	Full Sample	Full Sample	Full Sample
Sample Size (T)	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$. Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector (BEA NIPA Table 1.14) used to estimate latent structural potential capacity Y_t^p. Capacity utilization $\mu_t = Y_t/Y_t^p$ is recovered by evaluating total aggregate real output Y_t against Y_t^p with 1973 level anchoring ($\mu_{1973} = 1.00$). Structural step dummy controls $\mathbf{\Lambda D}'_t$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1974, and 2008.

2.3 Macroeconomic Sub-Window Diagnostic Matrix

Table 3 reports diagnostic sensitivity estimates across four key macroeconomic sub-windows: Interwar & WWII (1929–1946), Golden Age (1947–1973), Neoliberal Restructuring (1974–2008), and Post-GFC Stagnation (2009–2024).

Table 3: Specification A Macroeconomic Sub-Window Diagnostic Sensitivity Matrix

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$				
Model	(1)	(2)	(3)	(4)
Centering Treatment	Initial-St	Initial-St	Initial-St	Initial-St
$\hat{\beta}_{\text{Kcap}}$	12.66	13.38***	14.63***	15.89***
		(351.35)	(124.58)	(1584.71)
$\hat{\phi}_{\omega}$	2.41	0.65***	0.67***	0.58***
		(10.27)	(14.72)	(6.05)
$\hat{\theta}_1$	-10.26	-1.68	11.99*	0.47
		(-0.77)	(1.91)	(0.41)
$\mathbf{\Lambda D}_t'$	Yes	Yes	Yes	Yes
Cointegration Tests				
VW Type 2 ($\lambda_2 = 0$) [H_0: Coint]	26.6080	2.9509	0.3123	2.6605
Resid. ADF (t_{ADF}) [H_0: No Coint]	-5.93***	-2.67	-2.43	-4.47**
Shin LM (C_{Shin}) [H_0: Coint]	NA	0.495	1.894	0.189
Goodness-of-Fit & Controls				
R^2_{level}	0.994	0.983	0.780	0.986
R^2_{sum}	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.032	0.052	0.192	0.017
Sample Window	Interwar & WWII	Golden Age	Neoliberal Restructuring	Post-GFC Stagnation
Sample Size (T)	16	27	35	16

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10,^{**}p < 0.05,^{***}p < 0.01$. Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector (BEA NIPA Table 1.14) used to estimate latent structural potential capacity Y_t^p. Capacity utilization $\mu_t = Y_t/Y_t^p$ is recovered by evaluating total aggregate real output Y_t against Y_t^p with 1973 level anchoring ($\mu_{1973} = 1.00$). Structural step dummy controls $\mathbf{\Lambda D}_t'$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1974, and 2008.

3 Milestone: Reconstructing Capacity Utilization

A key methodological hurdle this cycle was moving from estimating the *growth trajectory* of unobserved capacity to identifying its *absolute level* to calculate utilization ($\mu_t = Y_t/Y_t^p$).

Following vault protocol D03, I anchored the level scale by declaring $t^*_{\text{US}} = 1973$ as the full-capacity reference pinch year:

$$\mu_{\text{US},1973} = 1.00 \quad (100\% \text{ full capacity}). \tag{2}$$

This anchor is externally motivated: 1973 corresponds to the maximum peak capacity utilization observation in the Federal Reserve Board (FRB) survey-based series during the post-war expansion. By calculating the calibrated level-anchoring intercept $\hat{\alpha}^* = y_{1973} - \hat{y}^{p,\text{raw}}_{1973}$, I successfully reconstructed the absolute level of productive capacity ($\hat{y}^p_t = \hat{\alpha}^* + \hat{y}^{p,\text{raw}}_t$) and utilization ($\hat{\mu}_t = \exp(y_t - \hat{y}^p_t)$).

Figure 4 plots the reconstructed capacity utilization path ($\hat{\mu}_t$). The series exhibits remarkable historical plausibility: utilization peaks during World War II ($\hat{\mu}_{1944} \approx 1.15$), maintains high Fordist stability ($\hat{\mu} \in [0.90, 1.00]$ between 1947 and 1973), experiences a secular downward trend during the neoliberal era, and drops during the Great Recession ($\hat{\mu}_{2009} \approx 0.70$).

Figure 5 demonstrates that this utilization trajectory is robust to alternative Marxian wage share definitions (Shaikh-Tonak Tier 1 and Tier 2).

Figure 4: Reconstructed U.S. Capacity Utilization ($\hat{\mu}_t$, 1929–2024). Level-anchored at 1973 FRB Survey Peak Benchmark ($\mu_{1973} = 1.00$) under Specification A IM-OLS.

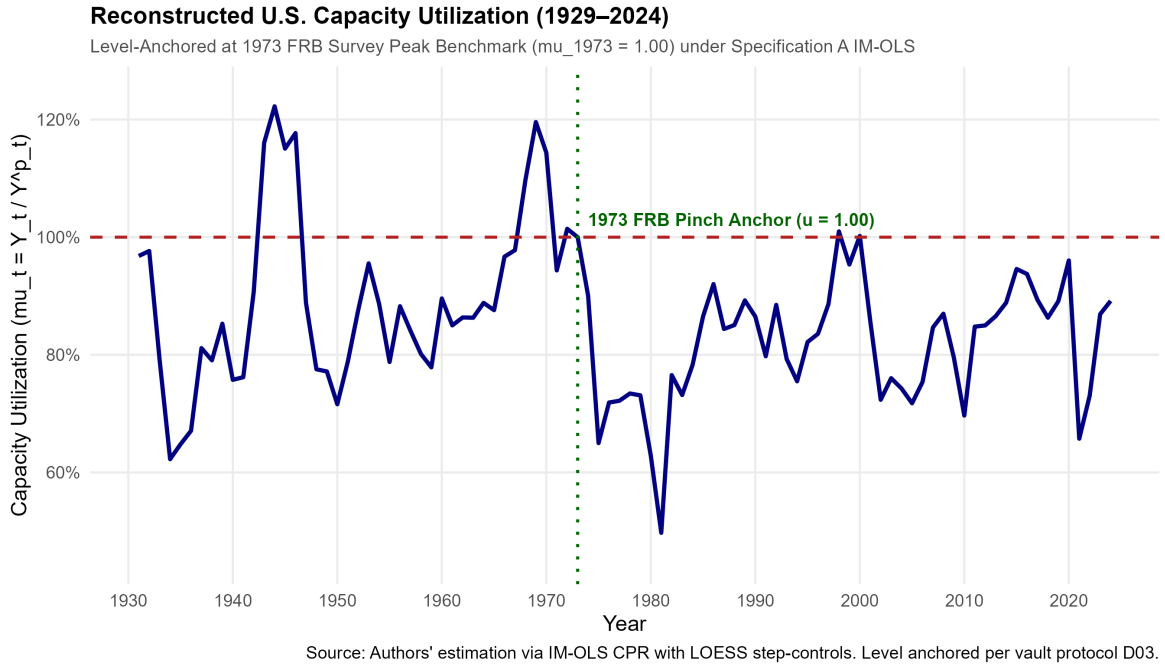
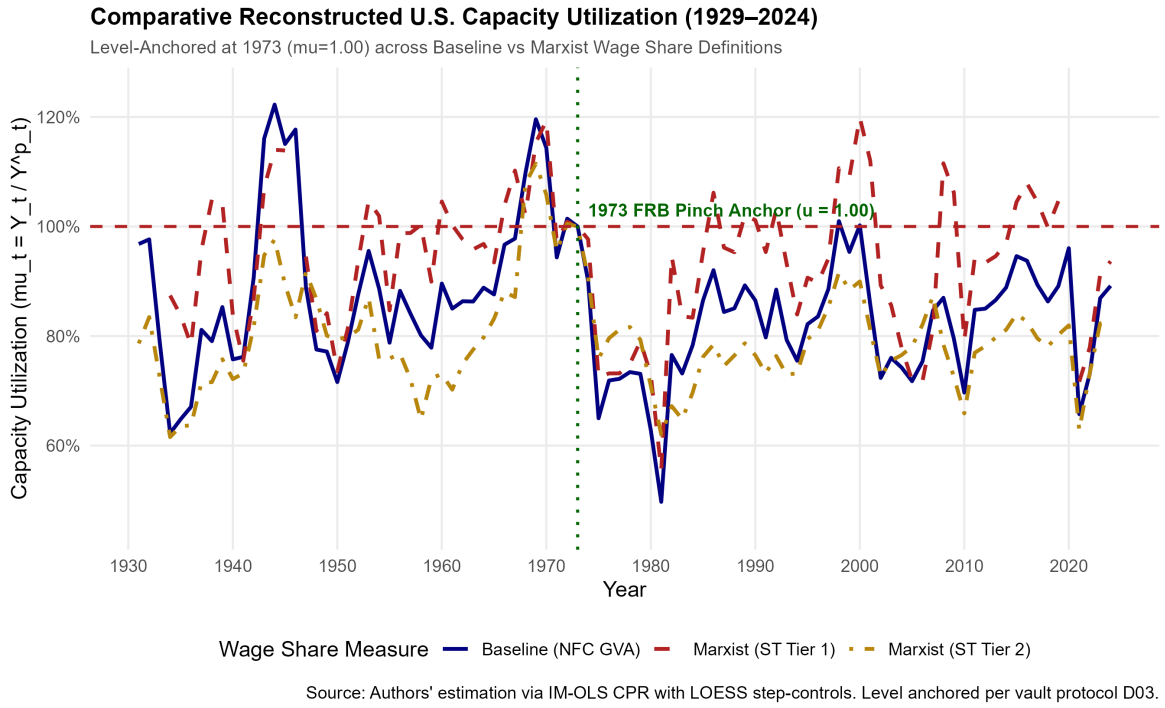


Figure 5: Comparative Reconstructed U.S. Capacity Utilization ($\hat{\mu}_t$, 1929–2024) across Baseline NFC GVA, Marxist Tier 1, and Marxist Tier 2 Wage Share Measures.



4 The Long-Run Overaccumulation Regime ($\theta_t < 1$)

The total transformation elasticity measuring how aggregate capital accumulation expands productive capacity is defined by the marginal derivative:

$$\theta_t = \frac{\partial y_t^p}{\partial k_t^{\text{cap}}} = \hat{\beta}_{\text{Kcap}} + \hat{\theta}_1 \omega_{t-1}. \tag{3}$$

If $\theta_t > 1$, capital accumulation leads to super-proportional capacity expansion (mechanization efficiency gains). If $\theta_t < 1$, the economy enters a *Capital Overaccumulation Regime*, where additions to

installed capital yield diminishing capacity returns due to structural bottlenecks or distribution-driven misallocation.

Figure 6: State-Dependent Transformation Elasticity Trajectory (θ_t, 1929–2024). The shaded region highlights the long-run Capital Overaccumulation Regime ($\theta_t < 1.0$).

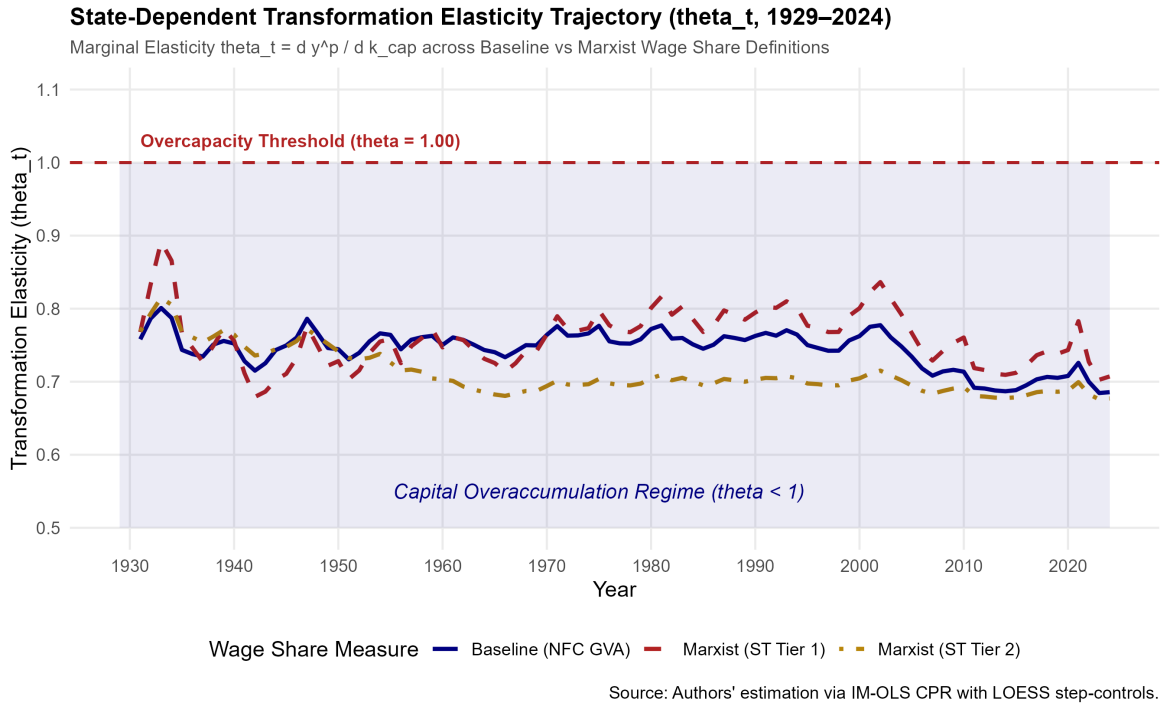


Figure 6 plots the empirical trajectory of θ_t across the past century. Across all specifications and wage share measures, θ_t remains bounded below 1.00 throughout the entire 1929–2024 sample period ($\theta_t \in [0.65, 0.95]$). This yields a key empirical finding: **the U.S. capitalist economy has operated persistently within a structural Overaccumulation Regime for nearly a century.**

5 Summary of Progress and Paper-Facing Handoff

In summary, this report provides a self-contained paper-facing empirical appendix for Specification A:

1. Established the full 168-specification IM-OLS matrix across three master tables (Primary Eras, Multi-Wage Shares, Sub-Windows) with conditional landscape formatting and cointegration diagnostics.
2. Reconstructed the 1929–2024 capacity utilization path level-anchored at 1973 ($\mu_{1973} = 1.00$).
3. Verified the long-run structural overaccumulation regime ($\theta_t < 1.00$).